API, Database Schemas, and Data Acquisition

Detection and Characterization of Subsurface Ice in Lunar South Polar Regions Using Chandrayaan-2 Radar and Imagery Data for Landing Site and Rover Traverse Planning

1. Data Acquisition Strategy

Primary Chandrayaan-2 data should be obtained from ISRO's Planetary Data System archive through ISSDC PRADAN and Chandrayaan-2 Map Browse. ISSDC states that Chandrayaan-2 data from CLASS, CHACE-2, XSM, IIRS, TMC-2, OHRC, DFRS, and DFSAR are processed by level definitions and provided in PDS standards for archival and dissemination.

Need	Instrument/Product	Where to Get It	Use in Project
Radar evidence for subsurface ice	Chandrayaan-2 DFSAR L/S-band SAR, full or hybrid polarimetry	PRADAN: https://pradan.issdc.gov.in/ch2/ and CH2 Map Browse	CPR, DOP, scattering behavior, radar feature stack
Landing hazard imagery	OHRC calibrated products	CH2 Map Browse: select South Pole projection and CH2_OHR_Calibrated_Product	Boulder/crater proxy, morphology, landing-site visual context
Terrain and DEM	TMC-2 stereo/DEM products where available	PRADAN/CH2 Map Browse	Slope, roughness, route cost, landing safety
Hydration/mineral context	IIRS hyperspectral products	PRADAN	Optional 3-micron hydration/mineral evidence layer
Geometry/navigation	SPICE kernels and footprints	PRADAN/ISSDC	Co-registration, illumination geometry, product provenance
Auxiliary validation	LRO LOLA slope/roughness/class products	NASA PGDA/PDS	Independent regional sanity check for TMC-2 terrain behavior
Production hardening	USGS ISIS and PDS validation workflows	USGS ISIS documentation	Future projection/calibration audit trail

2. Practical Download Steps

- Create/login to PRADAN if required: https://pradan.issdc.gov.in/ch2/.
- Open Chandrayaan-2 Map Browse at https://chmapbrowse.issdc.gov.in.
- Set map projection to South Pole for lunar south-polar AOIs.
- Use instrument footprint filters: DFSAR/SAR products for radar, CH2_OHR_Calibrated_Product for OHRC, and TMC-2 products for terrain context.
- Select footprints over target PSRs, inspect product metadata, and download PDS products.
- If browser download limits or timeouts occur, use PRADAN bulk download guidance and keep a local manifest of product IDs.

3. Current Downloaded Product Set

Payload	Product	Role
DFSAR	ch2_sar_ncls_20200913t042439405_d_fp_d18.zip	Radar evidence source for candidate subsurface-ice indicators.
TMC-2	ch2_tmc_ndn_20231203T0019079527_d_dtm_d18.zip	South-pole digital terrain model for slope, accessibility, illumination proxy, and traverse route.
TMC-2	ch2_tmc_ndn_20231203T0019079527_d_oth_d18.zip	Orthographic context layer; browse used for visual terrain context.
OHRC	ch2_ohr_ncp_20260103T0609041371_d_img_d18.zip	High-resolution optical hazard-context strip.
OHRC	ch2_ohr_ncp_20260103T1005176450_d_img_d18.zip	Second optical hazard-context strip for comparison.
NASA LOLA	PGDA LDRM_80S_1000MPP_ADJ_* products	Independent slope, roughness, and terrain-class validation crop.

4. Candidate API Endpoints

Method	Path	Purpose	Key Response
GET	/api/aoi	List demo AOIs	AOI id, bounds, centroid, available layers
POST	/api/aoi	Create custom AOI	AOI id and ingest status
GET	/api/layers/{aoi_id}	List raster/vector layers	Tile URLs, legends, resolution, source products
POST	/api/score/ice	Run or fetch ice-likelihood scoring	Raster summary, candidate targets, confidence
GET	/api/metrics/{layer_id}	Fetch dynamic metrics for a selected dashboard layer	Confidence, status label, candidate table, profile bars, processing chain
GET	/api/validation/lola/{aoi_id}	Return external LOLA comparison	TMC mean slope, LOLA mean slope, mean difference, caveats
POST	/api/score/landing	Rank landing ellipses	Safety/resource/science score breakdown
POST	/api/traverse	Plan route from site to targets	Path geometry, distance, cost, hazard events
GET	/api/demo/judge-mode	Return guided demo sequence	Step titles, target layer ids, narration, UI focus areas
GET	/api/products/{product_id}	Product provenance	Instrument, acquisition, processing, checksum
POST	/api/reports/mission-brief	Generate PDF report	Report URL and status

5. API Object Sketch

IceScoreRequest: aoi_id, layer_set_id, model_version, weights, thresholds, min_confidence. **IceCandidate**: id, geometry, score, confidence, area_m2, evidence: {cpr, dop, psr, morphology, roughness}, source_product_ids. **TraverseRequest**: aoi_id, start_site_id, target_ids, max_slope_deg, avoid_psr_mode, energy_budget_wh, science_weight. **LayerMetric**: dynamic metrics object with layer_id, plan_label, confidence, confidence_status, profile_bars, candidate_rows, processing_steps. **ValidationResult**: validation_id, source, compared_layer_ids, metrics_json, caveats.

6. Database Schema

Table	Core Fields	Purpose
raw_products	id, instrument, product_id, url, acquisition_time, level, footprint_geom, checksum, local_path	Immutable data provenance
aoi	id, name, bounds_geom, centroid, notes, created_at	Study regions and demo areas
raster_layers	id, aoi_id, product_id, layer_type, uri, crs, resolution_m, min_val, max_val, processing_version	COG/Zarr-derived map layers
feature_stats	id, aoi_id, layer_id, geom, mean, p05, p50, p95, nodata_pct	AOI and candidate summaries
validation_results	id, aoi_id, source_name, source_url, compared_layer_ids, metrics_json, caveats, created_at	External validation such as LOLA vs TMC-2 slope sanity checks
layer_metrics	id, layer_id, plan_label, confidence, status, bars_json, candidate_rows_json, processing_steps_json	Dynamic dashboard metric payloads
ice_candidates	id, aoi_id, geom, score, confidence, method, evidence_json, product_ids, created_at	Ranked ice-likelihood targets
landing_sites	id, aoi_id, geom, safety_score, science_score, resource_score, total_score, constraints_json	Candidate landing zones
traverse_paths	id, aoi_id, landing_site_id, target_id, geom, distance_m, total_cost, hazard_events_json	Rover routes
demo_steps	id, order_index, title, target_layer_id, narration, metric_focus_json	Judge Demo Mode story sequence
reports	id, aoi_id, type, uri, generated_at, config_json	Generated mission briefs

7. Current Evidence and LOLA Validation

External validation now uses NASA PGDA/LRO LOLA south-pole slope, roughness, and terrain-class products cropped to the Chandrayaan-2 TMC-2 AOI. The TMC-derived mean slope is 10.67 deg, the LOLA mean slope is 11.63 deg, and the mean difference is 0.96 deg. Because LOLA is 1000 m/pixel, this is a regional sanity check, not meter-scale hazard validation.

The prototype route from LZ-A to SCI-B is generated with A* over downsampled TMC-2 accessibility and cold-trap proxy rasters. The route has 50 path pixels and a relative cost of 151.74. It is a screening route for judge demonstration and needs geodetic calibration, obstacle inflation, and rover dynamics before mission use.

8. Source-to-Evidence Ledger

Source	Status	Contribution
ISRO PRADAN	Used	Official Chandrayaan-2 DFSAR, TMC-2, and OHRC product downloads.
Chandrayaan-2 Map Browse	Used	AOI/product discovery and south-pole footprint selection.
DFSAR science release	Science anchor	CPR/DOP interpretation target; current prototype uses ratio-based candidate evidence.
TMC-2 south-pole DTM/orthobrowse	Used	Slope, accessibility, hillshade, cold-trap proxy, and route cost layers.
OHRC	Context	High-resolution visual hazard context until exact footprint registration is complete.
NASA PGDA/LRO LOLA	Used for validation	Independent 1000 m/pixel slope, roughness, and class sanity check.
USGS ISIS	Roadmap	Production-grade planetary projection/calibration path documented for later hardening.

9. Spatial Indexing and Storage

- Use PostGIS geometry columns for AOIs, footprints, candidate regions, landing ellipses, and traverses.
- Store large rasters as files, not database blobs; keep URIs and metadata in PostgreSQL.
- Use ST_Intersects for product discovery by AOI and GIST indexes on all geometry fields.
- Use product checksums and processing_version to make every score reproducible.

Source Links

ISRO Science Data Archive - Chandrayaan-2 PRADAN: https://pradan.issdc.gov.in/ch2/

ISSDC Chandrayaan-2 mission data page: https://www.issdc.gov.in/chandrayaan2.html

ISRO DFSAR subsurface-ice release, May 27, 2026: https://www.isro.gov.in/Chandrayaan2_Dual_Frequency_Synthetic_Aperture_Radar.html

Chandrayaan-2 Map Browse: https://chmapbrowse.issdc.gov.in

VEDAS note on downloading Chandrayaan-2 OHRC products: https://vedas.sac.gov.in/static/pdf/SIH_2024/SIH1732_CH2_PS.pdf

NASA Planetary Data System Geosciences Node: https://pds-geosciences.wustl.edu/

NASA PGDA LRO LOLA south-pole products: https://pgda.gsfc.nasa.gov/products/90

USGS ISIS planetary image processing: https://isis.astrogeology.usgs.gov/